

11. Sequences and Series

11.1. Sequences

11.1 Learning Objectives

- Understand how to write a sequence and find their formula if they have one
- Find the limit of a sequence using limit laws and theorems
- Determine whether sequences converge or diverge using theorems
- Determine whether a sequence is increasing or decreasing, and whether it is bounded

Definition 11.1.1 Sequence

A **sequence** is a list of numbers $a_1, a_2, a_3, \dots, a_n, \dots$ in a definite order.

The a_i are called the **terms (or values)** of the sequence.

Consider the sequence

$$1, 1, 2, 3, 5, 8, 13, \dots$$

What is a_1 ?

$$\underbrace{1}_{a_1}, 1, 2, 3, 5, 8, 13, \dots$$

This is the first term in the list, so it is $a_1 = 1$.

What is a_6 ?

$$1, 1, 2, 3, 5, \underbrace{8}_{a_6}, 13, \dots$$

This is the sixth term in the list, so it is $a_6 = 8$.

Definition 11.1.2

The integer n is called the **index** of a_n . It indicates what term in the sequence you are considering.

We can think of the sequence $a_1, a_2, \dots, a_n, \dots$ as a function that sends the positive integer n to the n^{th} value in the sequence, a_n .

$$\begin{aligned} f(n) &= a_n, \text{ so } f(1) = a_1 \\ f(2) &= a_2 \\ f(3) &= a_3 \\ &\vdots \end{aligned}$$

The length of a sequence refers to the **number of terms** in the sequence. Sequences can be either infinite or finite.

Note

Notation: We typically denote sequences with some kind of parentheses or brackets.

$$(a_n)_{n=1}^{\infty}, [a_n]_{n=1}^{\infty}, \{a_n\}_{n=1}^{\infty}$$

all refer to the sequence

$$(a_1, a_2, a_3, \dots, a_n, \dots)$$

where a_i is the i^{th} term in the sequence.

Note: the indexing does not have to start at $n = 1$, and sometimes the index labeling is left off (something like just “ (a_n) ” for example).

Example 11.1.1

List the first 6 terms of the following sequences

1. $(a_n)_{n=1}^{\infty} = (\sqrt{n})_{n=1}^{\infty}$
2. $(b_n)_{n=3}^{\infty} = (\cos(\pi n))_{n=3}^{\infty}$

Solution

1. $(a_n)_{n=1}^{\infty} = (\sqrt{n})_{n=1}^{\infty}$

$$a_1 = 1, a_2 = \sqrt{2}, a_3 = \sqrt{3}, a_4 = 2, a_5 = \sqrt{5}, a_6 = \sqrt{6}$$

1. $(b_n)_{n=3}^{\infty} = (\cos(\pi n))_{n=3}^{\infty}$

$$b_3 = \cos(3\pi) = -1, b_4 = \cos(4\pi) = 1, b_5 = -1, b_6 = 1, b_7 = -1, b_8 = 1$$

Just because the indexing starts at $n = 3$, doesn't change how we read the first 6 terms.

Example 11.1.2

Write a rule that specifies the values of the sequence

$$(b_n)_{n=1}^{\infty} = \left(-\frac{1}{2}, \frac{2}{3}, -\frac{3}{4}, \frac{4}{5}, \dots\right).$$

Solution

Notice, the numerator is the index, and the denominator is one larger than the index. So, accounting for the alternating $\pm$ signs,

$$b_n = (-1)^n \frac{n}{n+1}.$$

Definition 11.1.3

A **recurrence relation** for a sequence $a_0, a_1, a_2, \dots$ is a formula that relates each value a_k to certain predecessors $a_{k-1}, a_{k-2}, \dots$.

Example 11.1.3

Define a sequence $c_0, c_1, c_2, \dots$ recursively as follows: For all integers $k \geq 2$,

1. $c_k = c_{k-1} + kc_{k-2} + 1$, we call this the recurrence relation
2. $c_0 = 1$ and $c_1 = 2$, we call this the initial conditions

Find c_2, c_3, c_4 .

Solution

$$c_2 = c_1 + 2 \cdot c_0 + 1 = 2 + 2(1) + 1 = 5$$

$$c_3 = c_2 + 3 \cdot c_1 + 1 = 5 + 3(2) + 1 = 12$$

$$c_4 = c_3 + 4 \cdot c_2 + 1 = 12 + 4(5) + 1 = 33$$

Definition 11.1.4

The sequence (a_n) **converges** to the number L if $\lim_{n \rightarrow \infty} a_n = L$.

If no such number L exists, we say that (a_n) **diverges**.

We say the sequence (a_n) **diverges to (minus) infinity** if $\lim_{n \rightarrow \infty} a_n = \pm\infty$

What does this mean?

If we go back to thinking about a sequence as a function that maps integers to real numbers, then we can essentially think of the terms in the sequence as being part of a continuous function with the same behavior as the sequence at integers. Then we can look at the limit of the function.

Theorem 11.1.1

If $\lim_{x \rightarrow \infty} a_n = L$ and $f(n) = a_n$ when n is an integer, then

$$\lim_{n \rightarrow \infty} a_n = L.$$

Example 11.1.4

Does the sequence $\left(\frac{n^2 + 3n + 1}{2n^2 + 1} \right)_{n=1}^{\infty}$ converge?

Solution

It suffices to check that $\lim_{x \rightarrow \infty} \frac{x^2 + 3x + 1}{2x^2 + 1}$ converges, so we can look at this and use regular limit techniques.

$$\begin{aligned}
 \lim_{x \rightarrow \infty} \frac{x^2 + 3x + 1}{2x^2 + 1} & \text{ has IFL } \frac{\infty}{\infty} \\
 & \stackrel{\text{LH}}{=} \lim_{x \rightarrow \infty} \frac{2x + 3}{4x} \text{ has IFL } \frac{\infty}{\infty} \\
 & \stackrel{\text{LH}}{=} \lim_{x \rightarrow \infty} \frac{2}{4} \\
 & = \frac{1}{2}
 \end{aligned}$$

Because the function where $f(n) = a_n$ converges to $\frac{1}{2}$, the sequence $(a_n)_{n=1}^{\infty} = \left(\frac{n^2 + 3n + 1}{2n^2 + 1} \right)_{n=1}^{\infty}$ converges to $\frac{1}{2}$ as well.

Theorem 11.1.2 Limit Laws

Let (a_n) and (b_n) be convergent sequences and let c be a constant. Then the following are true.

- $\lim_{n \rightarrow \infty} a_n \pm b_n = \lim_{n \rightarrow \infty} a_n \pm \lim_{n \rightarrow \infty} b_n$
- $\lim_{n \rightarrow \infty} ca_n = c \lim_{n \rightarrow \infty} a_n$
- $\lim_{n \rightarrow \infty} a_n b_n = \left(\lim_{n \rightarrow \infty} a_n \right) \left(\lim_{n \rightarrow \infty} b_n \right)$
- $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \frac{\lim_{n \rightarrow \infty} a_n}{\lim_{n \rightarrow \infty} b_n}$ assuming $\lim_{n \rightarrow \infty} b_n \neq 0$
- $\lim_{n \rightarrow \infty} (a_n)^p = \left(\lim_{n \rightarrow \infty} a_n \right)^p$ for $a_n > 0, p > 0$

Example 11.1.5

Decide whether the sequence is convergent or divergent. If the sequence is convergent, then find its limit.

$$((-1)^n)_{n=1}^{\infty}$$

Solution

$$((-1)^n)_{n=1}^{\infty} = (-1, 1, -1, 1, -1, 1, -1, 1, -1, 1, \dots)$$

This sequence **diverges** since the terms infinitely alternate between two different values, never getting closer to either.

Example 11.1.6

Decide whether the sequence is convergent or divergent. If the sequence is convergent, then find its limit.

$$\left(\frac{\ln(n)}{n}\right)_{n=1}^{\infty}$$

 Solution

Looking at the function $f(x) = \frac{\ln(x)}{x}$ the limit is

$$\lim_{x \rightarrow \infty} \frac{\ln(x)}{x} \stackrel{\text{LH}}{=} \lim_{x \rightarrow \infty} \frac{\frac{1}{x}}{1} = 0$$

So the sequence $(a_n)_{n=1}^{\infty} = \left(\frac{\ln(n)}{n}\right)_{n=1}^{\infty}$ is convergent, and it converges to 0.

 Recall

For some integer n the factorial of that integer, $n!$, is the product with it and every positive integer less than it. $n! = n \cdot (n-1) \cdot (n-2) \cdot \dots \cdot 2 \cdot 1$

examples:

- $5! = 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 = 120$
- $4! = 4 \cdot 3 \cdot 2 \cdot 1 = 24$
- $3! = 3 \cdot 2 \cdot 1 = 6$
- $2! = 2 \cdot 1 = 2$
- $1! = 1$
- We also define $0! = 1$ (this comes from combinatorics)

Example 11.1.7

Decide whether the sequence is convergent or divergent. If the sequence is convergent, then find its limit.

$$\left(\frac{(3n-1)!}{(3n+1)!}\right)_{n=1}^{\infty}$$

 Solution

$$\begin{aligned} \lim_{n \rightarrow \infty} \frac{(3n-1)!}{(3n+1)!} &= \lim_{n \rightarrow \infty} \frac{(3n-1)(3n-2)(3n-3)\dots 3 \cdot 2 \cdot 1}{(3n+1)(3n)(3n-1)(3n-2)(3n-3)\dots 3 \cdot 2 \cdot 1} \\ &= \lim_{n \rightarrow \infty} \frac{1}{(3n+1)(3n)} \\ &= 0 \end{aligned}$$

The sequence $\left(\frac{(3n-1)!}{(3n+1)!}\right)_{n=1}^{\infty}$ converges, and it converges to 0.

Example 11.1.8

Decide whether the sequence is convergent or divergent. If the sequence is convergent, then find its limit.

$$\left(\frac{3^n}{n^2}\right)_{n=1}^{\infty}$$

 **Solution**

$$\begin{aligned}\lim_{n \rightarrow \infty} \frac{3^n}{n^2} &\stackrel{\text{LH}}{=} \lim_{n \rightarrow \infty} \frac{\ln(3) \cdot 3^n}{2n} \\ &\stackrel{\text{LH}}{=} \lim_{n \rightarrow \infty} \frac{(\ln(3))^2 \cdot 3^n}{2} \\ &= \infty\end{aligned}$$

The sequence $\left(\frac{3^n}{n^2}\right)_{n=1}^{\infty}$ diverges to infinity.

11.1a Section Summary:

- We went over some of the basics of sequences and how to tell if they converge or diverge. We will continue this in the next day of notes.